Question 1

(a) True

Let (X,d) be a quasi-metric space and a pseudo-metric space. Let $x, y, z \in X$. Since (X,d) is a quasi-metric space, $d(x,y) = 0 \iff x = y$ and $d(x,z) \leq d(x,y) + d(y,z)$. Then since (X,d) is a pseudo-metric space, $d(x,y) = d(y,x)$ making (X,d) a metric space

(b) False

Consider the metric on $\mathbb{R}$ defined by

$$d(x,y) = \begin{cases} 0 & \text{if } x \leq y \\ 1 & \text{otherwise} \end{cases}$$

First we show this is indeed a quasi-metric space. Let $x, y \in \mathbb{R}$. Clearly if $x = y$, then $x \leq y$ so $d(x,y) = 0$ and if $d(x,y) = 0 = d(y,x)$, then $x \leq y$ and $y \leq x$ so $x = y$. Now let $z \in \mathbb{R}$. Then if $x \leq z$, $d(x,z) = 0 \leq 0 + 0 \leq d(x,y) + d(y,z)$. So now suppose $x > z$, then we cannot have $x \leq y$ and $y \leq z$ as this implies $x \leq z$, so $x > y$ or $y > z$ meaning $d(x,y) + d(y,z) \geq 1 = d(x,z)$ proving the triangle inequality. Now for sequence $(\frac{1}{n})$ in $\mathbb{R}$, $d(\frac{1}{n}, 2) = d(\frac{1}{n}, 3) = 0$ for all $n \in \mathbb{N}$ since $\frac{1}{n} \leq 2 \leq 3$. So $d(\frac{1}{n}, 2) \to 0$ and $d(\frac{1}{n}, 3) \to 0$ as $n \to \infty$ but $2 \neq 3$

(c) False

Consider the usual metric on $\mathbb{R}$ and sets $A_n = \{\frac{1}{n}\}$ for $n \in \mathbb{N}$. Each A_n is closed, since if a sequence is in A_n, it must be constant and thus converge to $\frac{1}{n} \in A_n$. But now $A = \bigcup_{n \in \mathbb{N}} A_n = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, ...\}$ which is not closed because sequence $(\frac{1}{n})$ is in A but converges to $0 \notin A$

(d) True

Let X be finite. Now let (y_n) be a convergent sequence in X, which converges to y. Now let $\epsilon = \min\{d(x,y) : x \in X, x \neq y\}$ (note this exists since X is finite, so there are finite metrics $d(x,y)$). Also, $\epsilon > 0$ since $\epsilon = 0 \Rightarrow \exists x \in X, d(x,y) = 0$ and $x \neq y$ which is a contradiction. Now if $y_n \neq y$, then $d(y_n, y) \geq \epsilon$. But since $y_n \to y$ as $n \to \infty$, there exists some $N_\epsilon \in \mathbb{N}$ such that for all $n \geq N_\epsilon$, $d(y_n, y) < \epsilon$. So $y_n = y$ for all $n \geq N_\epsilon$

(e) False. Take the usual metric on $\mathbb{R}$ with sets $(0,1)$ and $(1,2)$. Then $\overline{(0,1) \cap (1,2)} = \overline{\emptyset} = \emptyset$, but $\overline{(0,1)} \cap \overline{(1,2)} = [0,1] \cap [1,2] = \{1\} \neq \emptyset$

Question 2

Let $x, y, z \in X$. We have to show the separation property (M1), the symmetry property (M2), and the triangle inequality (M3)

(M1): First suppose $x = y$. (X, d) is a metric space so $d(x, y) = 0$ and thus $d^*(x, y) = \frac{d(x,y)}{1+d(x,y)} = \frac{0}{1+0} = 0$.
Conversely, if $d^*(x, y) = 0$, then $\frac{d(x,y)}{1+d(x,y)} = 0$. Since $d(x, y) \geq 0, 1 + d(x, y) \neq 0$. Then we get that $d(x, y) = 0 \Rightarrow x = y$
(M2): Note that $d^*(x, y) = \frac{d(x,y)}{1+d(x,y)} = \frac{d(y,x)}{1+d(y,x)} = d^*(y, x)$ [symmetry from (X, d)]
(M3): Function $f : [0, \infty) \to [0, \infty)$ defined by $f(t) = \frac{t}{1+t} = 1 - \frac{1}{1+t}$ is increasing because $0 \leq x \leq y \Rightarrow \frac{1}{x+1} \geq \frac{1}{y+1} \Rightarrow -\frac{1}{x+1} \leq -\frac{1}{y+1} \Rightarrow 1 - \frac{1}{x+1} \leq 1 - \frac{1}{y+1} \Rightarrow f(x) \leq f(y)$.
Now since (X, d) is a metric space, $d(x, z) \leq d(x, y) + d(x, z)$. Then $d(x, y) \in [0, \infty)$ for all $x, y \in X$ so $f(d(x, z)) = d^*(x, z) \leq f(d(x, y) + d(y, z)) = \frac{d(x,y)+d(y,z)}{1+d(x,y)+d(y,z)} = \frac{d(x,y)}{1+d(x,y)+d(y,z)} + \frac{d(y,z)}{1+d(x,y)+d(y,z)} \leq \frac{d(x,y)}{1+d(x,y)} + \frac{d(y,z)}{1+d(y,z)} = d^*(x, y) + d^*(y, z)$ as required, and thus (X, d^*) is a metric space